A Minor Project Synopsis

On

Plant Disease Detection

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in

Computer Science and Engineering

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Signature of Supervisor: _____

Date: 27/3/26

MINOR PROJECT SYNOPSIS

Session 2025 – 26

Plant Disease Detection Using Convolutional Neural Networks (CNN)

Bachelor of Technology – Computer Science & Engineering

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1 Introduction

Agriculture is the backbone of our economy, and healthy crops are essential for food security. However, plant diseases cause massive crop losses every year, affecting farmers financially and threatening food supply.

Traditionally, farmers rely on visual inspection or experts to detect diseases, which is slow, costly, and not always accurate. With the rise of Artificial Intelligence (AI) and Deep Learning, we can now build smart systems that **automatically detect plant diseases from leaf images** — faster and more accurately than the human eye.

This project uses **Convolutional Neural Networks (CNN)**, a type of deep learning model that is very good at understanding images, to identify whether a plant leaf is healthy or diseased, and if diseased, what disease it has.

2 Problem Statement

Farmers, especially in rural areas, often cannot identify plant diseases early enough. By the time the disease spreads, a large part of the crop is already damaged. The challenges include:

- Lack of agricultural experts in remote areas.
- Manual disease detection is time-consuming and error-prone.
- Existing chemical-based detection methods are expensive.
- Climate change is increasing the frequency of new plant diseases.

Our solution: An AI-powered application where a farmer simply **takes a photo of the affected leaf**, and the system instantly tells them what disease it is and how to treat it.

3 Literature Review

Researcher	Key Finding
Mohanty et al. (2016)	Used CNN with PlantVillage dataset; achieved 99.35% accuracy in a controlled lab setting.
Too et al. (2019)	Compared DenseNet, InceptionV4, and VGG models; found DenseNet gave the best accuracy.
Ramcharan et al. (2017)	Tested deep learning for cassava disease detection using smartphone images – showed real-world feasibility.
Ferentinos (2018)	Showed that transfer learning (using pre-trained models) drastically reduces training time.
Barbedo (2018)	Identified key challenges: lighting conditions, overlapping leaves, and background noise in real images.

4 Objectives

1. Build a CNN model that can identify plant leaf diseases from images.
2. Train the model on a large dataset (PlantVillage) with 38 different disease categories.
3. Achieve at least 90% accuracy on test data.
4. Create a simple user interface where users can upload a leaf image and get results.
5. Provide brief treatment suggestions along with disease detection.
6. Make the system accessible for real-world use (web or mobile app).

5 Methodology

Step-by-Step Process:

Step 1 – Data Collection	We use the PlantVillage dataset, which has over 54,000 images of healthy and diseased plant leaves across 38 classes. This is a well-known, trusted dataset used in agriculture research.
Step 2 – Data Preprocessing	All images are resized to 224×224 pixels. We apply data augmentation (flipping, rotating, zooming) to create more training samples and prevent overfitting. Pixel values are normalized to range [0, 1].
Step 3 – CNN Model Design	We build a CNN with multiple Convolutional Layers (to extract features like edges, colors, textures), MaxPooling Layers (to reduce size and focus on important features), Dropout Layers (to avoid memorizing training data), and a Dense Output Layer (to classify into 38 disease categories).
Step 4 – Transfer Learning	We also test pre-trained models like VGG16 and ResNet50, which are already trained on millions of images. This speeds up training significantly and improves accuracy.
Step 5 – Model Training	The model is trained using the Adam optimizer and Categorical Cross-Entropy loss function. Training runs for 20–50 epochs on GPU.
Step 6 – Evaluation	We test the model on unseen images and measure accuracy, precision, recall, and F1-score.
Step 7 – Deployment	The trained model is integrated into a Flask web application where users can upload an image and get instant disease predictions.

6 Tools & Technologies

Python 3.x	Main programming language — easy to use for AI/ML projects.
TensorFlow / Keras	Used to build and train the CNN model.
OpenCV	Used for image reading, resizing, and preprocessing.
NumPy & Pandas	For data handling and numerical computations.

Matplotlib	To plot accuracy/loss graphs and visualize results.
Flask	Python web framework to create the user-facing web application.
Google Colab / Jupyter	For running and testing the model with GPU support.
PlantVillage Dataset	Open-source dataset with 54,000+ plant disease images.
Git & GitHub	For version control and team collaboration.

7 Expected Outcomes

By the end of this project, we expect to deliver:

- A trained CNN model with accuracy above 90% on the PlantVillage test dataset.
- A working web application where anyone can upload a leaf photo and get a disease diagnosis.
- A clear comparison of our custom CNN vs. pre-trained models (VGG16, ResNet50).
- Graphs showing training accuracy, validation accuracy, and loss curves.
- A confusion matrix showing which diseases are detected most accurately.
- Disease treatment suggestions displayed alongside the diagnosis.

8 Applications

Farmers: Can take a photo with a smartphone and immediately know if their plant is diseased — no expert needed.

Agriculture Departments: Can monitor disease outbreaks across large areas quickly using AI-powered scanning.

Agri-tech Companies: Can integrate the model into drone systems to scan entire fields automatically.

Research Institutes: Can use the model to study disease patterns and develop better treatments.

Food Safety Agencies: Can check imported plants for diseases at borders before they spread.

9 Conclusion and Future Work

This project demonstrates how Artificial Intelligence can solve a real-world agricultural problem. By training a CNN on leaf images, we can automate disease detection — making it faster, cheaper, and accessible to everyone, even in remote areas without agricultural experts.

What we have done so far:

- *Studied existing research and identified gaps.*
- *Selected the PlantVillage dataset for training.*
- *Designed the CNN architecture.*
- *Set up the development environment.*

Future Scope:

- Extend to more plant species and uncommon diseases.
- Develop a mobile app (Android/iOS) for field use.
- Integrate with IoT sensors for real-time crop monitoring.
- Add a multilingual interface for regional farmers.
- Use federated learning to improve the model without sharing private data.

Team Leader Signature: _____	Date: _____

REFERENCES

1. Image Processing-Based Detection

Early approaches focused on traditional image processing techniques such as segmentation, color analysis, and texture extraction.

- **Reference:**
  Al-Hiary et al. (2011) – *"Fast and Accurate Detection and Classification of Plant Diseases"*
- **Explanation:**
  This study used image processing techniques to detect plant diseases by analyzing leaf color, shape, and texture. The system applied segmentation to isolate infected areas and then used classification algorithms. It was effective but limited in handling complex backgrounds and varying lighting conditions.

2. Machine Learning Approaches

With advancements in ML, classifiers like SVM, KNN, and Decision Trees were used.

- **Reference:**
  Pujari et al. (2016) – *"Classification of Fungal Disease Symptoms in Plants Using Machine Learning Techniques"*
- **Explanation:**
  The researchers extracted features such as color and texture and applied ML algorithms like Support Vector Machine (SVM). Results showed improved accuracy compared to traditional methods, but performance depended heavily on feature extraction quality.

3. Deep Learning-Based Detection

Deep learning models, especially Convolutional Neural Networks (CNNs), significantly improved accuracy.

- **Reference:**
  Mohanty et al. (2016) – *"Using Deep Learning for Image-Based Plant Disease Detection"*
- **Explanation:**
  This work used CNN models like AlexNet and GoogleNet on the PlantVillage dataset. It achieved accuracy above 99%, showing that deep learning can automatically learn features without manual extraction.

4. Transfer Learning Techniques

To improve performance with limited data, transfer learning became popular.

- **Reference:**
  Too et al. (2019) – *"A Comparative Study of Fine-Tuning Deep Learning Models for Plant Disease Identification"*
- **Explanation:**
  This study compared pre-trained models such as VGG16, ResNet, and Inception. Fine-tuning these models improved accuracy and reduced training time, making them suitable for real-world applications.

5. Mobile and Real-Time Detection Systems

Recent research focuses on deploying models in real-time systems and mobile applications.

- **Reference:**
  Ferentinos (2018) – *"Deep Learning Models for Plant Disease Detection and Diagnosis"*
- **Explanation:**
  The study developed a system capable of detecting multiple plant diseases using deep CNNs. It achieved high accuracy and demonstrated practical usability in agriculture.

6. Hybrid and IoT-Based Systems

Some studies combine AI with IoT for smart agriculture.

- **Reference:**
  Brahimi et al. (2017) – *"Deep Learning for Tomato Diseases: Classification and Symptoms Visualization"*
- **Explanation:**
  This work used CNNs along with visualization techniques to identify disease regions. It helped farmers understand disease spread and take preventive actions.